

Lab: Accessing MongoDB using Python

Accessing MongoDB using Python



Objectives

After completing this lab you will be able to:

- Access the MongoDB database from Python with the pymongo driver
- Perform basic operations such as selecting, inserting and listing using Python
- Create a Python program to run the MongoDB operations

Exercise 1 - Install the pymongo driver

We need the pymongo driver to be installed in order to access the mongodb database from Python.

Run the below command on the terminal.

```
1 python3 -m pip install pymongo
```

This installs the Python mongodb driver as in the image below.

```
theia@theia:~/home/project$ python3 -m pip install pymongo
Collecting pymongo
  Downloading https://files.pythonhosted.org/packages/10/3b/46541b4ee3000019b8ef5b1847292ddc77f492c162bc4d49c424db7fc97a/pymongo-4.1.1-cp36-cp36m-manylinux1_x86_64.whl (464kB)
    100% |#####| 471kB 2.7MB/s
Installing collected packages: pymongo
Successfully installed pymongo-4.1.1
```

SKILLS NET...WelcomeMongoDB xMySQL

DATABASES

BIG DATA

CLOUD

EMBEDDABLE AI

OTHER

Launch Applica...

MongoDBACTIVE

3.6.30.54.02.0.2

Connect to MongoDB and mongo-express directly in your Skills Network Labs environment.

CreateDelete

SummaryConnection InformationDetails

MONGO_USERNAME:

root

MONGO_HOST:

172.21.249.202

MONGO_PORT:

27017

URL:

https://labs-mongo-echoing-swift-chef.mongo.databases.labs.skills.network

MONGO_URL:

https://labs-mongo-echoing-swift-chef.mongo.databases.labs.skills.network

MongoDB CLI Command:

mongosh mongodb://root:suFen8T0Q7P101fDAjaXjHac@172.21.249.202:27017

Exercise 2 - Start the server

Open the MongoDB database page by clicking the button below:

 [Open MongoDB Page in IDE](#)

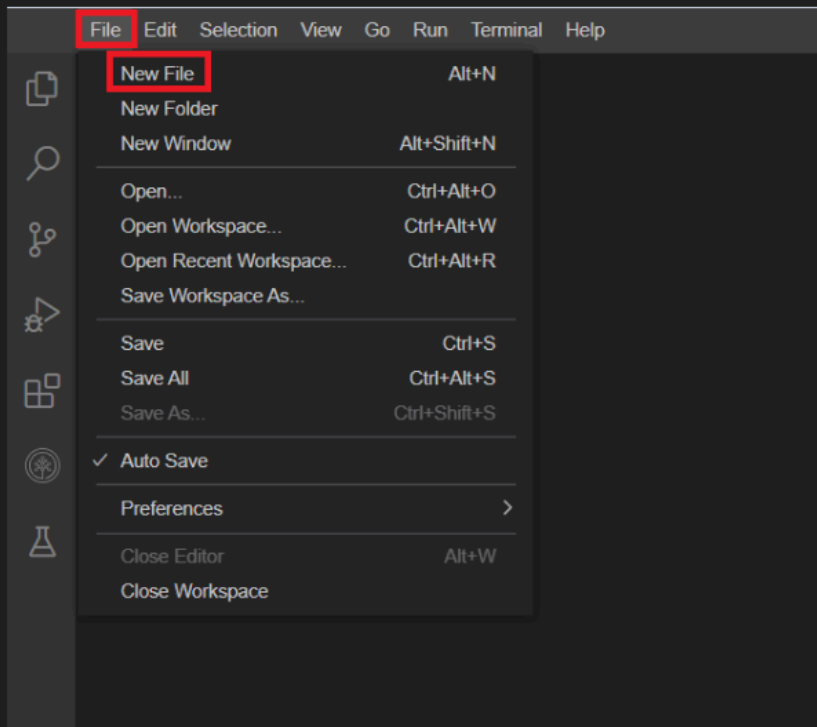
On that page, click the **Create** button to create a MongoDB database.

Note down the user name and password from the **Connection Information** tab.

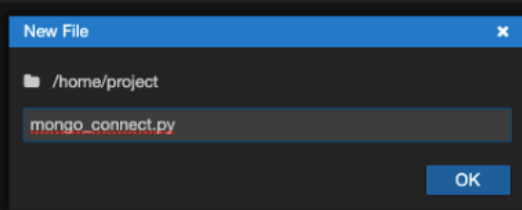


Exercise 3 - Connect to mongodb server using Python

On the menu, use **File->New File** to create a new file, as in the image below.



Give the name as **mongo_connect.py**, as in the figure below, and click on OK.

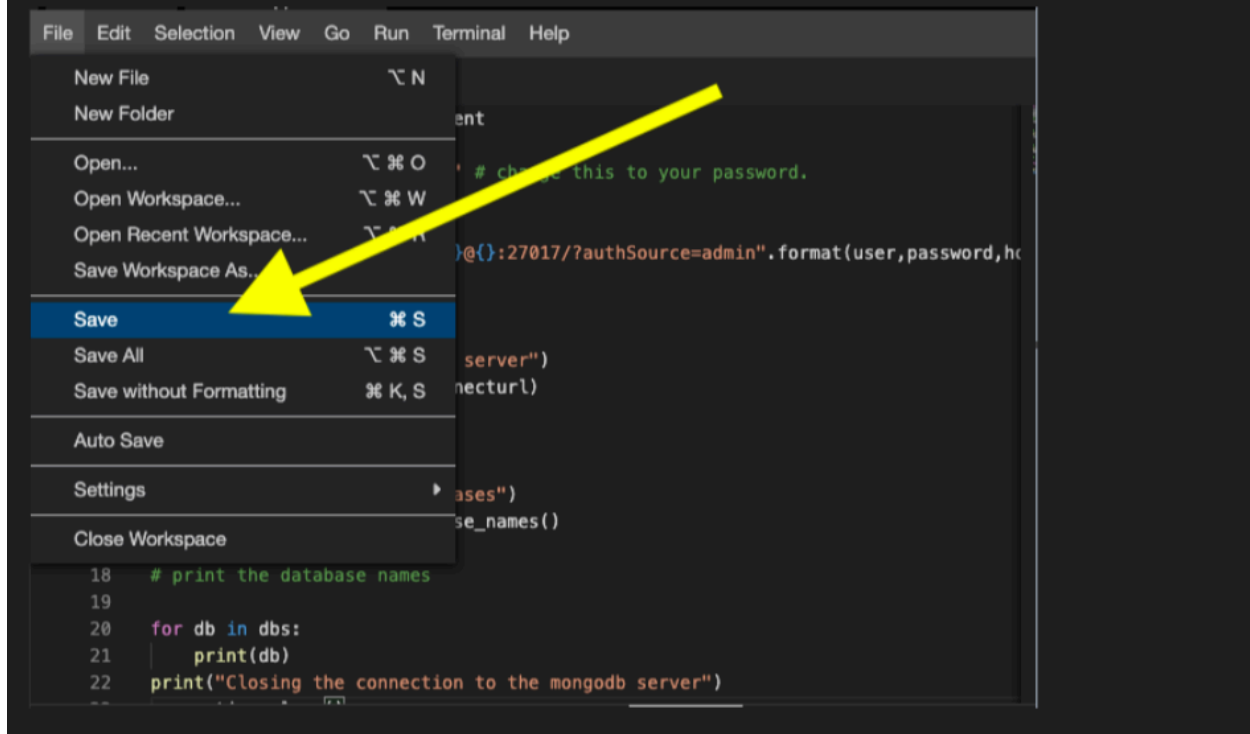


Copy and paste the below code into the newly opened file.

```
1  from pymongo import MongoClient
2  user = 'root'
3  password = 'MjQwOTgtcnNhbm5h' # CHANGE THIS TO THE PASSWORD YOU NOTED IN THE EARLIER EXERCISE - 2
4  host='mongo'
5  #create the connection url
6  connecturl = "mongodb://{}:{}@{}:27017/?authSource=admin".format(user,password,host)
7
8  # connect to mongodb server
9  print("Connecting to mongodb server")
10 connection = MongoClient(connecturl)
11
12 # get database list
13 print("Getting list of databases")
14 dbs = connection.list_database_names()
15
16 # print the database names
17
18 for db in dbs:
19     print(db)
20 print("Closing the connection to the mongodb server")
21
22 # close the server connection
23 connection.close()
```

PLEASE ENSURE THAT YOU HAVE REPLACED THE PASSWORD VALUE IN THE FILE ABOVE WITH THE PASSWORD FOR YOUR MONGODB SERVER THAT YOU COPIED AFTER IT WAS STARTED.

Save the code file using the **File->Save** menu option as in the image below.



Copy and paste the below code on the terminal to execute this file.

```
1 python3 mongo_connect.py
```

You should see an output like the one in the image below.

```
theia@theiadocker-rsannareddy:/home/project$ python mongo_connect.py
Connecting to mongodb server
Getting list of databases
admin
config
local
Closing the connection to the mongodb server
theia@theiadocker-rsannareddy:/home/project$
```

MONGO_URL:

`https://labs-mongo-echoing-swift-chef.mongo.databases.labs.skills.network`

MongoDB CLI Command:

`mongosh mongodb://root:suFen8T0Q7P101fDAjaXjHac@172.21.249.202:27017`

MONGO_COMMAND:

`mongosh mongodb://root:suFen8T0Q7P101fDAjaXjHac@172.21.249.202:27017`

MONGO_PASSWORD:

`suFen8T0Q7P101fDAjaXjHac`

MONGO_TITLE:

`MongoDB Database`

MONGO_ID:

`labs-mongo-echoing-swift-chef`

Exercise 4 - Working with documents

In this exercise we will make the Python program do the following tasks:

- connect to the mongodb server.
- select a database named **training**.
- select a collection named **python**.
- insert a sample document.
- query all the documents in the **training** database and **python** collection.
- close the connection to the server.

Open a new file named mongo_query.py.

Copy and paste the below code into mongo_query.py.

```
1  from pymongo import MongoClient
2  user = 'root'
3  password = 'MjQwOTgtcnNhbm5h' # CHANGE THIS TO THE PASSWORD YOU NOTED IN THE EARLIER EXERCISE - 2
4  host='mongo'
5  #create the connection url
6  connecturl = "mongodb://{}:{@{}:27017/?authSource=admin".format(user,password,host)
7
8  # connect to mongodb server
9  print("Connecting to mongodb server")
10 connection = MongoClient(connecturl)
11
12 # select the 'training' database
13
14 db = connection.training
15
16 # select the 'python' collection
17
18 collection = db.python
19
20 # create a sample document
21
22 doc = {"lab":"Accessing mongodb using python", "Subject":"No SQL Databases"}
23
24 # insert a sample document
25
26 print("Inserting a document into collection.")
27 db.collection.insert_one(doc)
```

```
29 # query for all documents in 'training' database and 'python' collection
30
31 docs = db.collection.find()
32
33 print("Printing the documents in the collection.")
34
35 for document in docs:
36     print(document)
37
38 # close the server connecton
39 print("Closing the connection.")
40 connection.close()
```

PLEASE ENSURE THAT YOU HAVE REPLACED THE PASSWORD VALUE IN THE FILE ABOVE WITH THE PASSWORD FOR YOUR MONGODB SERVER THAT YOU COPIED AFTER IT WAS STARTED.

Save the file.

Run the file using the below command.

```
1  python3 mongo_query.py
```

Practice exercise

Write a Python program that can:

- connect to the mongodb server.
- select a database named **training**.
- select a collection named **mongodb_glossary**.
- insert the following documents into the collection **mongodb_glossary**.

```
{"database": "a database contains collections"}
```

```
{"collection": "a collection stores the documents"}
```

```
{"document": "a document contains the data in the form of key value pairs."}
```

- query and print all the documents in the **training** database and **mongodb_glossary** collection.
- close the connection to the server.

Solution to Practice exercise

```
1  from pymongo import MongoClient
2  user = 'root'
3  password = 'MjQwOTgtcnNhbm5h' # CHANGE THIS TO THE PASSWORD YOU NOTED IN THE EARLIER EXERCISE - 2
4  host='mongo'
5  #create the connection url
6  connecturl = "mongodb://{}:{:}@{:}:27017/?authSource=admin".format(user,password,host)
7
8  # connect to mongodb server
9  print("Connecting to mongodb server")
10 connection = MongoClient(connecturl)
11
12 # select the 'training' database
13
14 db = connection.training
15
16 # select the 'python' collection
17
18 collection = db.mongodb_glossary
19
20 # create documents
21
22 doc1 = {"database": "a database contains collections"}
23 doc2 = {"collection": "a collection stores the documents"}
24 doc3 = {"document": "a document contains the data in the form or key value pairs."}
25
26 # insert documents
27 print("Inserting documents into collection.")
28
29 db.collection.insert_one(doc1)
30 db.collection.insert_one(doc2)
31 db.collection.insert_one(doc3)
32
```



```
32
33 # query for all documents in 'training' database and 'python' collection
34
35 docs = db.collection.find()
36
37 print("Printing the documents in the collection.")
38
39 for document in docs:
40     print(document)
41
42 # close the server connecton
43 print("Closing the connection.")
44 connection.close()
```